

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum -590014, Karnataka.



LAB REPORT

On

ANALYSIS AND DESIGN OF ALGORITHMS (23CS4PCADA)

Submitted by

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**in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING**

in

COMPUTER SCIENCE AND ENGINEERING



B.M.S. COLLEGE OF ENGINEERING

**(Autonomous Institution under
VTU) BENGALURU-560019**

February to June 2026

CERTIFICATE

**B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering**



This is to certify that the Lab work entitled “**ANALYSIS AND DESIGN OF ALGORITHMS**” carried out by Manshi Kumari (**1BF24CS167**), who is bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2025-26. The Lab report has been approved as it satisfies the academic requirements in respect of Analysis and Design of Algorithms Lab - (**23CS4PCADA**) work prescribed for the said degree.

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Course outcomes:

CO1	Analyze time complexity of algorithms using asymptotic notations.
CO2	Apply various algorithm design techniques for the given problem.
CO3	Evaluate the appropriate algorithm/design technique for solving a given problem.
CO4	Conduct practical experiments by applying appropriate algorithm design technique to solve problems.

Lab program 1:

Write program to obtain the Topological ordering of vertices in a given digraph.

```
#include <stdio.h>

#define MAX 100

int graph[MAX][MAX];
int visited[MAX];
int stack[MAX];
int top = -1;

void dfs(int v, int n) {

    visited[v] = 1;

    for(int i = 0; i < n; i++) {

        if(graph[v][i] && !visited[i]) {
            dfs(i, n);
        }
    }

    stack[++top] = v;
}

void topologicalSort(int n) {

    for(int i = 0; i < n; i++) {

        if(!visited[i]) {
            dfs(i, n);
        }
    }
    printf("Topological Ordering:\n");

    while(top != -1) {

        printf("%d ", stack[top--]);
    }
}
```

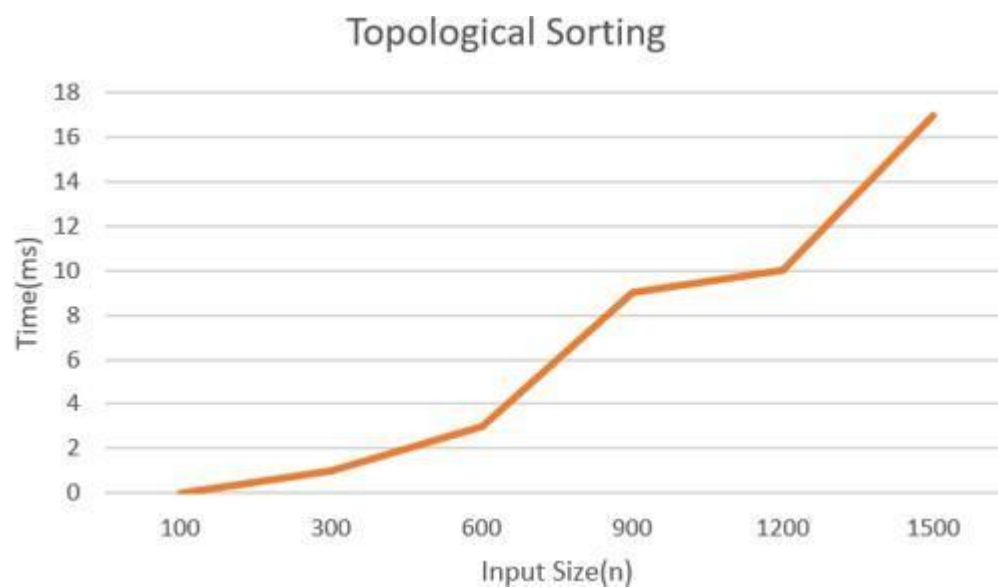


```
int main() {  
  
    int n, e;  
  
    printf("Enter number of vertices: ");  
    scanf("%d", &n);  
  
    printf("Enter number of edges: ");  
    scanf("%d", &e);  
  
    printf("Enter edges (u v):\n");  
  
    for(int i = 0; i < e; i++) {  
  
        int u, v;  
  
        scanf("%d %d", &u, &v);  
  
        graph[u][v] = 1;  
    }  
  
    topologicalSort(n);  
  
    return 0;  
}
```

Output:

```
Enter number of vertices: 3
Enter adjacency matrix:
0 1 0
0 0 1
1 0 0

Graph contains a cycle. Topological ordering not possible.
Process returned 0 (0x0)   execution time : 12.094 s
Press any key to continue.
```



Leetcode: 207. Course Schedule

```
#include <stdbool.h>
#include <stdlib.h>

bool canFinish(int numCourses, int** prerequisites, int prerequisitesSize, int*
prerequisitesColSize) {

    int indegree[numCourses];

    // Initialize indegree to 0
    for(int i = 0; i < numCourses; i++)
        { indegree[i] = 0;
        }

    // Create adjacency matrix
    int adj[numCourses][numCourses];

    for(int i = 0; i < numCourses; i++) {
        for(int j = 0; j < numCourses; j++) {
            adj[i][j] = 0;
        }
    }

    // Build graph and indegree
    for(int i = 0; i < prerequisitesSize; i++) {
        int a = prerequisites[i][0];
        int b = prerequisites[i][1];

        adj[b][a] = 1;
        indegree[a]++;
    }

    // Queue
    int queue[numCourses];
    int front = 0, rear = 0;

    // Push nodes with indegree 0
    for(int i = 0; i < numCourses; i++)
    {
        if(indegree[i] == 0) {
            queue[rear++] = i;
        }
    }
    int count = 0;
    while(front < rear) {
        int node = queue[front++];
```

```

count++;

for(int i = 0; i < numCourses; i++) {
    if(adj[node][i] == 1) {
        indegree[i]--;

        if(indegree[i] == 0) {
            queue[rear++] = i;
        }
    }
}

return count == numCourses;
}
}

```

Output:

The screenshot displays a submission page for a problem. The top section shows the submission status as 'Accepted' with 54/54 testcases passed, submitted on May 2. Below this, the 'Runtime' section indicates a time of 40 ms, which is 26.05% better than other submissions. The 'Memory' section shows a usage of 110.35 MB, which is 18.02% better. A performance graph shows the distribution of runtime times, with a peak around 40 ms. The 'Code' section on the right shows the C++ code used for the submission. The 'Testcase' section shows the input for Case 1: numCourses = 2 and prerequisites = [[1,0]].

Accepted 54 / 54 testcases passed
submitted at May 2

Runtime
40 ms | Beats 26.05%

Memory
110.35 MB | Beats 18.02%

Code

```

1 #include <stdio.h>
2 #include <stdlib.h>

```

Testcase Case 1 Case 2 +

numCourses =
2

prerequisites =
[[1,0]]

Lab program 2:

Implement Johnson Trotter algorithm to generate permutations

```
#include <stdio.h>

#define LEFT 0
#define RIGHT 1

void print(int a[], int dir[], int n)
{
    for(int i = 0; i < n; i++)
    {
        if(dir[a[i]] == LEFT)
            printf("%d<- ",
                a[i]);
        else
            printf("%d-> ", a[i]);
    }
    printf("\n");
}

int getMobile(int a[], int dir[], int n)
{
    int mobile = 0;

    for(int i = 0; i < n; i++)
    {
        if(dir[a[i]] == LEFT && i > 0)
        {
            if(a[i] > a[i-1] && a[i] > mobile)
                mobile = a[i];
        }

        if(dir[a[i]] == RIGHT && i < n-1)
        {
            if(a[i] > a[i+1] && a[i] > mobile)
                mobile = a[i];
        }
    }
    return mobile;
}
```

```
int findPos(int a[], int n, int mobile)
```

```

{
    for(int i = 0; i < n; i++)
    {
        if(a[i] == mobile)
            return i;
    }
    return -1;
}

```

```

void johnsonTrotter(int n)
{
    int a[n], dir[n+1];

    for(int i = 0; i < n; i++)
    {
        a[i] = i + 1;
        dir[i + 1] = LEFT;
    }
    print(a, dir, n);
    while(1)
    {
        int mobile = getMobile(a, dir, n);

        if(mobile ==
            0) break;
        int pos = findPos(a, n, mobile);

        if(dir[mobile] == LEFT)

        {
            int temp = a[pos];
            a[pos] = a[pos - 1];
            a[pos - 1] = temp;
            pos = pos - 1;
        }
        else
        {
            int temp = a[pos];
            a[pos] = a[pos + 1];
            a[pos + 1] = temp;

```

```

        pos = pos + 1;
    }

    for(int i = 1; i <= n; i++)
    {
        if(i > mobile)
            dir[i] = !dir[i];
    }

    print(a, dir, n);
}

int main()
{
    int n;

    printf("Enter n: ");
    scanf("%d", &n);

    johnsonTrotter(n);

    ;

    return 0;
}

```

Output:

```

Enter n: 3
1<- 2<- 3<-
1<- 3<- 2<-
3<- 1<- 2<-
3-> 2<- 1<-
2<- 3-> 1<-
2<- 1<- 3->

```


Lab program 3:

Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and record the time taken to sort.

```
#include <stdio.h>
#include
<stdlib.h>
#include <time.h>
void merge(int arr[], int lb, int mid, int ub)
{
    int n1 = mid - lb + 1;
    int n2 = ub - mid;

    int L[n1], R[n2];

    for (int i = 0; i < n1; i++)
        L[i] = arr[lb + i];

    for (int j = 0; j < n2; j++)
        R[j] = arr[mid + 1 + j];
    int i = 0, j = 0, k = lb;

    while (i < n1 && j < n2)
    {
        if (L[i] <= R[j])
            arr[k++] = L[i++];
        else
            arr[k++] = R[j++];
    }

    while (i < n1)
        arr[k++] = L[i++];

    while (j < n2)
        arr[k++] = R[j++];
}

void mergeSort(int arr[], int lb, int ub)
{
    if (lb < ub)
    {
```

```
int mid = (lb + ub) / 2;
```

```
        mergeSort(arr, lb, mid);
        mergeSort(arr, mid + 1, ub);

        merge(arr, lb, mid, ub);
    }
}

int main()
{
    int n;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n];
    printf("Enter elements:\n");
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    clock_t start = clock();

    mergeSort(arr, 0, n - 1);

    clock_t end = clock();

    double time_taken = (double)(end - start) / CLOCKS_PER_SEC;

    printf("\nSorted array:\n");

    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);

    printf("\n");

    printf("Time taken: %.6f seconds\n", time_taken);

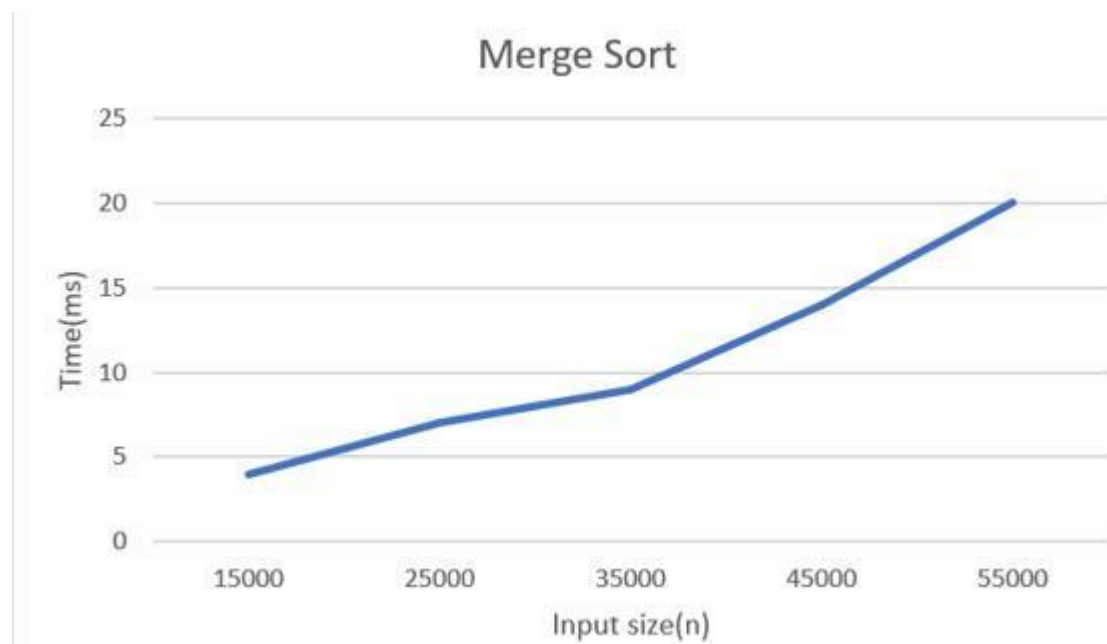
    return 0;
}
```

Output:

```
Enter number of elements: 5
Enter elements:
29 5 67 3 54

Sorted array:
3 5 29 54 67
Time taken: 0.000000 seconds

Process returned 0 (0x0)   execution time : 10.999 s
Press any key to continue.
|
```



Leetcode 912: Sort an array

```
#include <stdlib.h>

void merge(int nums[], int left, int mid, int right)
{
    int n1 = mid - left + 1;
    int n2 = right - mid;

    int L[n1], R[n2];

    for(int i = 0; i < n1; i++)
        L[i] = nums[left + i];

    for(int j = 0; j < n2; j++)
        R[j] = nums[mid + 1 + j];
    int i = 0, j = 0, k = left;

    while(i < n1 && j <

n2)

    {
        if(L[i] <= R[j])
        {
            nums[k] = L[i];
            i++;
        }
        else
        {
            nums[k] = R[j];
            j++;
        }
        k++;
    }

    while(i < n1)
    {
        nums[k] = L[i];
        i++;
    }
```

```
k++;  
}
```

```
while(j < n2)
{
    nums[k] = R[j];
    j++;
    k++;
}
}

void mergesort(int nums[], int left, int right)
{
    if(left < right)
    {
        int mid = left + (right - left) / 2;

        mergesort(nums, left, mid);
        mergesort(nums, mid + 1, right);

        merge(nums, left, mid, right);
    }
}

int* sortArray(int* nums, int numsSize, int* returnSize)
{
    mergesort(nums, 0, numsSize - 1);

    *returnSize = numsSize;

    return nums;
}
```

Output:

DescriptionAccepted xEditorialSolution

All Submissions

Accepted 54 / 54 testcases passed

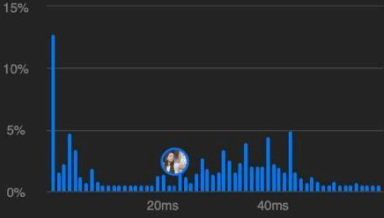
ManshiKu... submitted at Jun 02, 2026 22:55

Runtime

22 ms | Beats 64.76%

Memory

110.31 MB | Beats 18.02%



Code | C

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
```

Display a menu

Code

```
50 free(graph);
51 free(graphSize);
52 free(indegree);
53 free(queue);
54
55 return visited == numCourses;
56 }
57
```

SavedLn 1, Col 1

Testcase | Test Result

Case 1Case 2+

numCourses =

2

prerequisites =

[[1,0]]

</> Source ?

Lab program 4:

Sort a given set of N integer elements using Quick Sort technique and compute its time taken.

```
#include <stdio.h>
#include
<stdlib.h>
#include <time.h>

void swap(int *a, int *b)
{
    int temp = *a;
    *a = *b;
    *b = temp;
}

int partition(int arr[], int low, int high)
{
    int pivot = arr[high];
    int i = (low - 1);

    for (int j = low; j < high; j++)
    {
        if (arr[j] <= pivot)
        {
            i++;
            swap(&arr[i], &arr[j]);
        }
    }

    swap(&arr[i + 1], &arr[high]);
    return (i + 1);
}

void quickSort(int arr[], int low, int high)
{
    if (low < high)
    {
        int pi = partition(arr, low, high);

        quickSort(arr, low, pi - 1);
        quickSort(arr, pi + 1, high);
    }
}
```



```
int main()
{
    int N;

    printf("Enter number of elements: ");
    scanf("%d", &N);

    int arr[N];
    printf("Enter the elements:\n");

    for (int i = 0; i < N; i++)

    {
        scanf("%d", &arr[i]);
    }

    clock_t start, end;

    start = clock();

    quickSort(arr, 0, N - 1);

    end = clock();

    printf("Sorted array:\n");

    for (int i = 0; i < N; i++)
    {
        printf("%d ", arr[i]);
    }

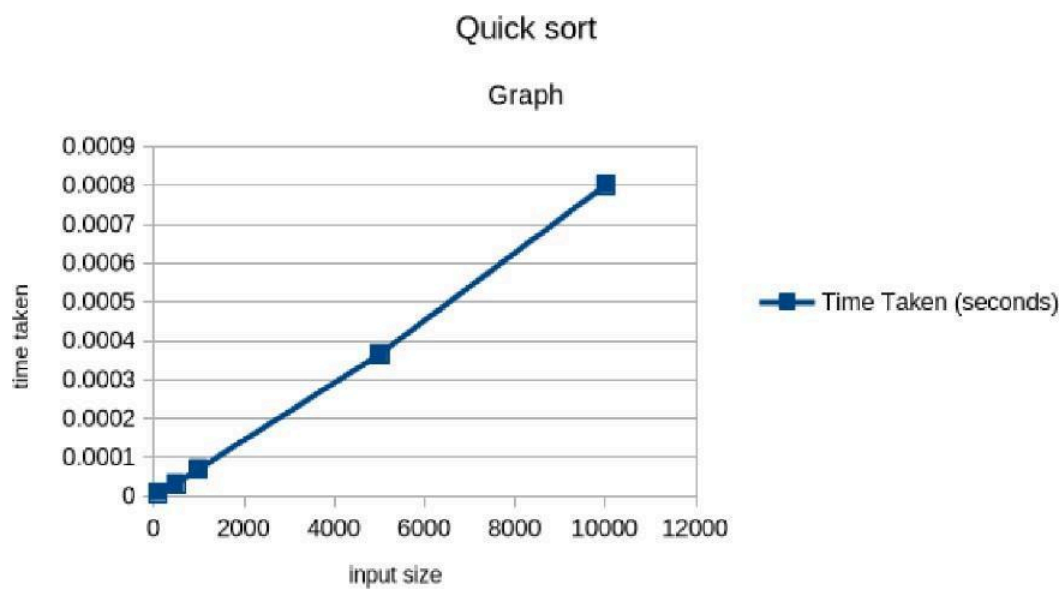
    double time_taken = ((double)(end - start)) / CLOCKS_PER_SEC;

    printf("\nTime taken: %f seconds\n", time_taken);

    return 0;
}
```


Output:

```
Enter number of elements: 8
Enter the elements:
10 3 7 2 9 1 5 8
Sorted array:
1 2 3 5 7 8 9 10
Time taken: 0.000000 seconds
```



Leetcode 207: Course Schedule

```
#include <stdio.h>
#include <stdlib.h>

// Function to check if you can finish all courses
int canFinish(int numCourses, int** prerequisites, int prerequisitesSize, int*
prerequisitesColSize) {
    // Build graph adjacency list
    int* indegree = (int*)calloc(numCourses, sizeof(int));
    int** graph = (int**)malloc(numCourses * sizeof(int*));
    int* graphSize = (int*)calloc(numCourses, sizeof(int));

    for (int i = 0; i < numCourses; i++) {
        graph[i] = (int*)malloc(numCourses * sizeof(int)); // worst case
    }

    for (int i = 0; i < prerequisitesSize; i++) {
        int course = prerequisites[i][0];
        int prereq = prerequisites[i][1];
        graph[prereq][graphSize[prereq]++] = course;
        indegree[course]++;
    }

    // Queue for BFS
    int* queue = (int*)malloc(numCourses * sizeof(int));
    int front = 0, rear = 0;

    // Add courses with indegree 0
    for (int i = 0; i < numCourses; i++) {
        if (indegree[i] == 0) {
            queue[rear++] = i;
        }
    }

    int visited = 0;
    while (front < rear) {
        int node = queue[front++];
        visited++;
        for (int i = 0; i < graphSize[node]; i++)
            { int neighbor = graph[node][i];
              indegree[neighbor]--;
              if (indegree[neighbor] == 0) {
                  queue[rear++] = neighbor;
              }
            }
    }
}
```



```

    }
}
}

// Free memory
for (int i = 0; i < numCourses; i++) {
    free(graph[i]);
}
free(graph);
free(graphSize);
free(indegree);
free(queue);

return visited == numCourses;
}

```

Output:

The screenshot displays a LeetCode submission interface. On the left, the submission status is 'Accepted' with 54/54 testcases passed. The user 'ManshiKu...' submitted it on Jun 02, 2026 at 23:00. Performance metrics show a runtime of 39 ms (30.55% beats) and memory usage of 110.49 MB (16.01% beats). A runtime distribution graph is shown below these metrics. The right side of the interface shows the C++ code and a test case input.

Code:

```

29     queue[rear++] = 1;
30 }
31 }
32
33 int visited = 0;
34 while (front < rear) {
35     int node = queue[front++];
36     visited++;
37     for (int i = 0; i < graphSize
38         [node]; i++) {
39         int neighbor = graph[node]
40             [i];
41         ...

```

Testcase:

Case 1 Case 2 +

numCourses =

2

prerequisites =

[[1,0]]

Source ?

Lab program 5:

Sort a given set of N integer elements using Heap Sort technique and compute its time taken.

```
#include <stdio.h>
```

```
void swap(int *a, int *b)
{
    int temp = *a;
    *a = *b;
    *b = temp;
}
```

```
void siftUp(int a[], int i)
{
    while(i > 0)
    {
        int parent = (i - 1) / 2;

        if(a[parent] < a[i])
        {
            swap(&a[parent],
                &a[i]); i = parent;
        }
        else
        {
            break;
        }
    }
}
```

```
void buildHeap(int a[], int n)
{
    for(int i = 1; i < n; i++)
    {
        siftUp(a, i);
    }
}
```

```
void heapify(int a[], int n, int i)
{
    int largest = i;
    int left = 2 * i + 1;
```

```

int right = 2 * i + 2;

if(left < n && a[left] > a[largest])
    largest = left;

if(right < n && a[right] > a[largest])
    largest = right;

if(largest != i)
{
    swap(&a[i], &a[largest]);
    heapify(a, n, largest);
}
}

void heapSort(int a[], int n)
{
    buildHeap(a, n);

    for(int i = n - 1; i > 0; i--)
    {
        swap(&a[0], &a[i]);
        heapify(a, i, 0);
    }
}

void print(int a[], int n)
{
    for(int i = 0; i < n; i++)
        printf("%d ", a[i]);

    printf("\n");
}

int main()
{
    int n;

    printf("Enter n: ");
    scanf("%d", &n);

    int a[n];

```

```
printf("Enter elements:\n");

for(int i = 0; i < n; i++)
    scanf("%d", &a[i]);

heapSort(a, n);

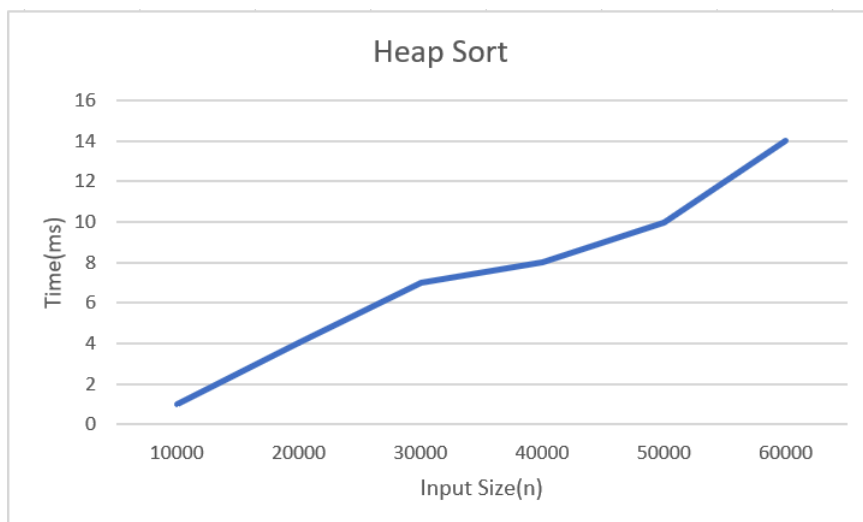
printf("Sorted array:\n");
print(a, n);

return 0;
}
```

Output:

```
Enter n: 6
Enter elements:
6
10
5
80
25
90
Sorted array:
5 6 10 25 80 90

Process returned 0 (0x0)   execution time : 49.457 s
Press any key to continue.
```



Lab program 6:

Implement 0/1 Knapsack problem using dynamic programming.

```
#include <stdio.h>

int knapsack(int W, int wt[], int val[], int n)
{
    int dp[n+1][W+1];

    for (int i = 0; i <= n; i++)
    {
        for (int w = 0; w <= W; w++)
        {
            if (i == 0 || w == 0)
            {
                dp[i][w] = 0;
            }
            else if (wt[i-1] <= w)
            {
                int include = val[i-1] + dp[i-1][w - wt[i-1]];
                int exclude = dp[i-1][w];

                dp[i][w] = (include > exclude) ? include : exclude;
            }
            else
            {
                dp[i][w] = dp[i-1][w];
            }
        }
    }
}

int selected[n];

for (int i = 0; i < n; i++)
    selected[i] = 0;

int i = n;
int w = W;

while (i > 0 && w > 0)
{
```

```

        if (dp[i][w] != dp[i-1][w])
        {
            selected[i-1] = 1;
            w = w - wt[i-1];
        }

        i--;
    }
    printf("\nItems included in Knapsack:\n");
    for (int i = 0; i < n; i++)
    {
        printf("%d ", selected[i]);
    }

    printf("\n");

    return dp[n][W];
}

int main()
{
    int n, W;

    printf("Enter number of items: ");
    scanf("%d", &n);

    int val[n], wt[n];

    printf("Enter profit of items:\n");
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &val[i]);
    }

    printf("Enter weights of items:\n");
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &wt[i]);
    }

    printf("Enter knapsack capacity: ");

```

```
scanf("%d", &W);

int result = knapsack(W, wt, val, n);

printf("\nMaximum value in Knapsack = %d\n",

result);

return 0;
}
```

Output:

```
Enter number of items: 5
Enter profit of items:
25 20 15 40 50
Enter weights of items:
3 2 1 4 5
Enter knapsack capacity: 6

Items included in Knapsack:
0 0 1 0 1

Maximum value in Knapsack = 65
```


Leetcode : 518. Coin Change II

```
int change(int amount, int* coins, int coinsSize) {  
  
    unsigned long long dp[amount + 1];  
  
    for(int i = 0; i <= amount; i++){  
        dp[i] = 0;  
    }  
  
    dp[0] = 1;  
  
    for(int i = 0; i < coinsSize; i++){  
        for(int j = coins[i]; j <= amount; j++){  
            dp[j] = dp[j] + dp[j - coins[i]];  
        }  
    }  
  
    return (int)dp[amount];  
}
```

Output:

The screenshot displays the LeetCode submission interface for problem 518. The left panel shows the submission status as 'Accepted' with 31/31 test cases passed. The user 'ManshiKu...' submitted the solution on Jun 02, 2026 at 23:02. The runtime is 7 ms, beating 87.23% of other submissions. The memory usage is 8.48 MB, beating 100.00%. A bar chart illustrates the distribution of runtime performance. The right panel shows the C++ code used for the solution, which implements a dynamic programming approach. The test case input is amount = 5 and coins = [1, 2, 5].

Code:

```
10  
11     for(int i = 0; i < coinsSize; i++){  
12         for(int j = coins[i]; j <= amount; j++){  
13             dp[j] = dp[j] + dp[j - coins[i]];  
14         }  
15     }  
16  
17     return (int)dp[amount];  
18 }
```

Testcase:

Case 1 Case 2 Case 3 +

amount =
5

coins =
[1, 2, 5]

Lab program 7:

Implement All Pair Shortest paths problem using Floyd's algorithm.

```
#include <stdio.h>

#define INF 99999
#define MAX 100

void printSolution(int dist[MAX][MAX], int V)
{
    printf("\nShortest distance matrix:\n");

    for (int i = 0; i < V; i++)
    {
        for (int j = 0; j < V; j++)
        {
            if (dist[i][j] == INF)
                printf("%7s", "INF");
            else
                printf("%7d", dist[i][j]);
        }
        printf("\n");
    }
}

void floydWarshall(int graph[MAX][MAX], int V)
{
    int dist[MAX][MAX];

    for (int i = 0; i < V; i++)
        for (int j = 0; j < V; j++)
            dist[i][j] = graph[i][j];

    for (int k = 0; k < V; k++)
    {
        for (int i = 0; i < V; i++)
        {
            for (int j = 0; j < V; j++)
            {
                if (dist[i][k] < INF &&
                    dist[k][j] < INF &&
                    dist[i][k] + dist[k][j] < dist[i][j])
                {

```

```

        dist[i][j] = dist[i][k] + dist[k][j];
    }
}
}
}

for (int i = 0; i < V; i++)
{
    if (dist[i][i] < 0)
    {
        printf("\nNegative cycle detected!\n");
        return;
    }
}

printSolution(dist, V);
}

int main()
{
    int V;
    int graph[MAX][MAX];

    printf("Enter number of vertices:
"); scanf("%d", &V);
    printf("\nEnter adjacency matrix (-1 for no edge):\n");

    for (int i = 0; i < V; i++)

    {
        for (int j = 0; j < V; j++)
        {
            scanf("%d", &graph[i][j]);

            if (graph[i][j] == -1)
                graph[i][j] = INF;
        }
    }

    for (int i = 0; i < V; i++)
        graph[i][i] = 0;

    floydWarshall(graph, V);

```



```
    return 0;  
}
```

Output:

```
Enter number of vertices: 4  
  
Enter adjacency matrix (-1 for no edge):  
0 3 -1 5  
2 0 -1 4  
-1 6 0 -1  
-1 -1 2 0  
  
Shortest distance matrix:  
    0    3    7    5  
    2    0    6    4  
    8    6    0   10  
   10    8    2    0  
  
Process returned 0 (0x0)   execution time : 30.376 s  
Press any key to continue.  
|
```

Leetcode 1334: Find the city with the smallest no of neighbors at a threshold distance

```
#include <limits.h>

int findTheCity(int n, int** edges, int edgesSize, int* edgesColSize, int distanceThreshold) {

    int dist[101][101];

    // Initialize matrix
    for(int i = 0; i < n;
    i++){
        for(int j = 0; j < n;
        j++){ if(i == j)
            dist[i][j] =
            0; else
            dist[i][j] = INT_MAX / 2;
        }
    }

    // Fill edge weights
    for(int i = 0; i < edgesSize; i++){
        int u = edges[i][0];
        int v = edges[i][1];
        int w =
        edges[i][2];

        dist[u][v] = w;
        dist[v][u] = w;
    }

    // Floyd Warshall
    for(int k = 0; k < n; k++){
        for(int i = 0; i < n;
        i++){
            for(int j = 0; j < n; j++){
                if(dist[i][j] > dist[i][k] + dist[k][j]){
                    dist[i][j] = dist[i][k] + dist[k][j];
                }
            }
        }
    }
}
```

}

}

```

int city = -1;
int minCount = INT_MAX;

// Count reachable
cities for(int i = 0; i <
n; i++){
    int count = 0;

    for(int j = 0; j < n; j++){
        if(i != j && dist[i][j] <= distanceThreshold){
            count++;
        }
    }

    if(count <=
        minCount){
        minCount = count;
        city = i;
    }
}

return city;
}

```

Output:

The screenshot displays a code submission interface with the following components:

- Submission Status:** Accepted (54 / 54 testcases passed). Submitted by ManshiKu... on Jun 02, 2026 23:03.
- Runtime Performance:** 9 ms, Beats 87.80%.
- Memory Performance:** 11.81 MB, Beats 10.66%.
- Performance Chart:** A bar chart showing the distribution of runtime times across various percentiles (3ms, 8ms, 13ms, 18ms).
- Code Editor:** Displays the C code for finding the city with the minimum count of reachable cities within a distance threshold.
- Test Results:** Shows two test cases, both passed (Case 1 and Case 2).
- Input Details:**
 - n = 4
 - edges = [[0, 1, 3], [1, 2, 1], [1, 3, 4], [2, 3, 1]]
 - distanceThreshold =

Lab program 8a:

Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.

```
#include <stdio.h>
#include <limits.h>

int minKey(int key[], int mstSet[], int V)
{
    int min = INT_MAX, min_index = -1;

    for (int v = 0; v < V; v++)
        if (mstSet[v] == 0 && key[v] < min) min = key[v], min_index = v;

    return min_index;
}

void primMST(int graph[50][50], int V)
{
    int parent[V], key[V], mstSet[V];
    int totalCost = 0;

    for (int i = 0; i < V; i++)
    {
        key[i] = INT_MAX;
        mstSet[i] = 0;
    }

    key[0] = 0;
    parent[0] = -1;

    for (int count = 0; count < V - 1; count++)
    {
        int u = minKey(key, mstSet, V);
        mstSet[u] = 1;

        for (int v = 0; v < V; v++)
        {
            if (graph[u][v] && mstSet[v] == 0 && graph[u][v] < key[v])
            {
```



```

        parent[v] = u;
        key[v] = graph[u][v];
    }
}
}
printf("\nEdges in MST:\n");

for (int i = 1; i < V; i++)

{
    printf("%d - %d : %d\n", parent[i], i, graph[i][parent[i]]);
    totalCost += graph[i][parent[i]];
}

printf("Total cost = %d\n", totalCost);
}

int main()
{
    int V;
    int graph[50][50];

    printf("Enter number of vertices: ");
    scanf("%d", &V);
    printf("Enter adjacency matrix (enter 0 if no edge, otherwise enter cost):\n");

    for (int i = 0; i < V; i++)

        for (int j = 0; j < V; j++)
            scanf("%d", &graph[i][j]);

    primMST(graph, V);

    return 0;
}

```


Output:

```
Enter number of vertices: 4
Enter adjacency matrix:
0 2 0 6
2 0 3 8
0 3 0 0
6 8 0 0
Edges in MST:
0 - 1 = 2
1 - 2 = 3
0 - 3 = 6
Minimum Cost = 11
```

Lab program 8b:

Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.

```
#include <stdio.h>
#include <stdlib.h>

struct Edge
{
    int src, dest, weight;
};

struct Subset
{
    int parent;
};

int find(struct Subset subsets[], int i)
{
    while (subsets[i].parent != i)
        i = subsets[i].parent;

    return i;
}

void Union(struct Subset subsets[], int x, int y)
{
    int xroot = find(subsets, x);
    int yroot = find(subsets, y);

    subsets[xroot].parent = yroot;
}

int compare(const void *a, const void *b)
{
    return ((struct Edge *)a)->weight - ((struct Edge *)b)->weight;
}

void KruskalMST(struct Edge edges[], int V, int E)
{
    struct Edge result[V];
```

```

struct Subset subsets[V];
qsort(edges, E, sizeof(struct Edge), compare);

for (int i = 0; i < V; i++)

    subsets[i].parent = i;

int e = 0;
int i = 0;

while (e < V - 1 && i < E)
{
    struct Edge next = edges[i++];

    int x = find(subsets, next.src);
    int y = find(subsets, next.dest);

    if (x != y)
    {
        result[e++] = next;
        Union(subsets, x, y);
    }
}

printf("\nEdges in MST:\n");

int total = 0;

for (i = 0; i < e; i++)
{
    printf("%d -- %d == %d\n",
           result[i].src,
           result[i].dest,
           result[i].weight);

    total += result[i].weight;
}

printf("Total Cost = %d\n", total);
}

int main()
{

```



```
int V;

printf("Enter number of vertices: ");
scanf("%d", &V);

int E = V * (V - 1) / 2;

struct Edge edges[E];

printf("Enter edges (src dest weight):\n");

for (int i = 0; i < E; i++)
{
    scanf("%d %d %d",
        &edges[i].src,
        &edges[i].dest,
        &edges[i].weight);
}

KruskalMST(edges, V, E);

return 0;
}
```

Output:

```
Enter number of vertices: 4
Enter adjacency matrix:
0 2 0 6
2 0 3 8
0 3 0 1
6 8 1 0
Edges in MST:
2 - 3 = 1
0 - 1 = 2
1 - 2 = 3
Minimum Cost = 6

Process returned 0 (0x0)    execution time : 26.320 s
Press any key to continue.
```

Lab program 9:

Implement Fractional Knapsack using Greedy technique

```
#include <stdio.h>
struct Item
{
    int weight;
    int value;
};

int main()
{
    int n, capacity;

    printf("Enter number of items: ");
    scanf("%d", &n);
    struct Item items[n];

    for(int i = 0; i < n; i++)
    {
        printf("Enter weight and value of item %d: ", i + 1);
        scanf("%d %d", &items[i].weight, &items[i].value);
    }

    printf("Enter knapsack capacity: ");
    scanf("%d", &capacity);

    double ratio[n];

    for(int i = 0; i < n; i++)
        ratio[i] = (double)items[i].value / items[i].weight;

    for(int i = 0; i < n - 1; i++)
    {
        for(int j = i + 1; j < n; j++)
        {
            if(ratio[i] < ratio[j])
            {
                double tempR = ratio[i];
```

```

        ratio[i] = ratio[j];

        ratio[j] = tempR;

        struct Item temp = items[i];
        items[i] = items[j];
        items[j] = temp;
    }
}
}

double maxVal = 0.0;
int remaining = capacity;

for(int i = 0; i < n; i++)
{
    if(items[i].weight <= remaining)
    {
        remaining -= items[i].weight;
        maxVal += items[i].value;
    }
    else
    {
        maxVal += ratio[i] * remaining;
        break;
    }
}

printf("Maximum value in Knapsack = %.2f\n", maxVal);

return 0;
}

```

Output:

```

Enter number of items: 7
Enter weight and value of item 1: 1 5
Enter weight and value of item 2: 3 10
Enter weight and value of item 3: 5 15
Enter weight and value of item 4: 4 7
Enter weight and value of item 5: 1 8
Enter weight and value of item 6: 3 9
Enter weight and value of item 7: 2 4
Enter knapsack capacity: 15
Maximum value in Knapsack = 51.00

```


Leetcode 455:Assign Cookies

```
#include <stdlib.h>
```

```
int compare(const void *a, const void *b) {  
    return (*(int *)a - *(int *)b);  
}
```

```
int findContentChildren(int* g, int gSize, int* s, int sSize) {
```

```
    // Sort greed factors and cookie sizes
```

```
    qsort(g, gSize, sizeof(int), compare);
```

```
    qsort(s, sSize, sizeof(int), compare);
```

```
    int i = 0; // children pointer
```

```
    int j = 0; // cookies pointer
```

```
    while (i < gSize && j < sSize) {
```

```
        if (s[j] >= g[i]) {
```

```
            i++;
```

```
        }
```

```
        j++;
```

```
    }
```

```
    return i;
```

```
}455
```

Output:

← All Submissions

Accepted 25 / 25 testcases passed

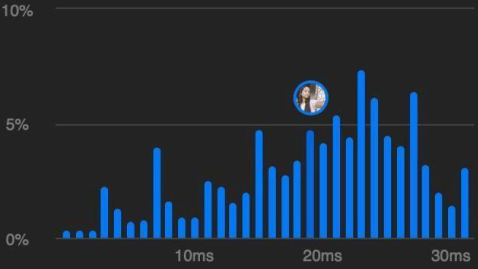
ManshiKu... submitted at Jun 02, 2026 23:09

Runtime

19 ms | Beats 63.93%

Memory

12.63 MB | Beats 38.15%



Code | C

```
1 #include <stdlib.h>
2
3 int compare(const void *a, const void *b) {
    return (b - a);
}
```

C Auto

```
15
16 while (i < gSize && j < sSize) {
17     if (s[j] >= g[i]) {
18         i++;
19     }
20     j++;
21 }
22
23 return i;
24 }
```

Saved Ln 18, Col 18

Testcase | Test Result

Case 1 Case 2 +

g =

[1,2,3]

s =

[1,1]

</> Source

Lab program 10:

From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.

```
#include <stdio.h>
#include <limits.h>

#define V 10

int minDistance(int dist[], int sptSet[], int vertices)
{
    int min = INT_MAX, min_index;

    for (int v = 0; v < vertices; v++)
    {
        if (sptSet[v] == 0 && dist[v] <= min)
        {
            min = dist[v];
            min_index = v;
        }
    }

    return min_index;
}

void printSolution(int dist[], int vertices, int src)
{
    printf("Vertex \t Distance from Source %d\n", src);

    for (int i = 0; i < vertices; i++)
    {
        printf("%d \t %d\n", i, dist[i]);
    }
}

void dijkstra(int graph[V][V], int src, int vertices)
{
    int dist[vertices];
    int
    sptSet[vertices];

    for (int i = 0; i < vertices; i++)
    {
```



```

        dist[i] = INT_MAX;
        sptSet[i] = 0;
    }

    dist[src] = 0;

    for (int count = 0; count < vertices - 1; count++)
    {
        int u = minDistance(dist, sptSet, vertices);
        sptSet[u] = 1;

        for (int v = 0; v < vertices; v++)
        {
            if (!sptSet[v] &&
                graph[u][v] &&
                dist[u] != INT_MAX &&
                dist[u] + graph[u][v] < dist[v])
            {
                dist[v] = dist[u] + graph[u][v];
            }
        }
    }

    printSolution(dist, vertices, src);
}

int main()
{
    int vertices;

    printf("Enter number of vertices: ");
    scanf("%d", &vertices);

    int graph[V][V];
    printf("Enter adjacency matrix (0 if no edge):\n");

    for (int i = 0; i < vertices; i++)

    {
        for (int j = 0; j < vertices; j++)
        {
            scanf("%d", &graph[i][j]);
        }
    }
}

```



```
}

int src;

printf("Enter source vertex: ");
scanf("%d", &src);

dijkstra(graph, src, vertices);

return 0;
}
```

Output:

```
Enter number of vertices: 5
Enter adjacency matrix (0 if no edge):
0 10 0 5 0
0 0 1 2 0
0 0 0 0 4
0 3 9 0 2
7 0 6 0 0
Enter source vertex: 0
Vertex    Distance from Source 0
0         0
1         8
2         9
3         5
4         7
```

Lab program 11:

Implement “N-Queens Problem” using Backtracking.

```
#include <stdio.h>
#include <stdbool.h>
#define MAX 20

int N;

int board[MAX][MAX];

bool isSafe(int row, int col)
{
    for(int i = 0; i < row; i++)
    {
        if(board[i][col])
            return false;

        if(col - (row - i) >= 0 &&
            board[i][col - (row - i)]) return false;

        if(col + (row - i) < N &&
            board[i][col + (row - i)])
            return false;
    }

    return true;
}

void printBoard()
{
    for(int i = 0; i < N; i++)
    {
        for(int j = 0; j < N; j++)
            printf("%c ", board[i][j] ? 'Q' : '.');

        printf("\n");
    }

    printf("\n");
}
```



```

}

void solve(int row)
{
    if(row == N)
    {
        printBoard();
        return;
    }

    for(int col = 0; col < N; col++)
    {
        if(isSafe(row, col))
        {
            board[row][col] = 1;

            solve(row + 1);

            board[row][col] = 0;
        }
    }
}

int main()
{
    printf("Enter the size of the board (N):
    "); scanf("%d", &N);

    if(N > MAX || N < 1)
    {
        printf("Invalid board size. Please enter between 1 and %d.\n", MAX);
        return 1;
    }

    solve(0);

    return 0;
}

```

Output:

```
Enter the size of the board (N): 4
. Q . .
. . . Q
Q . . .
. . Q .

. . Q .
Q . . .
. . . Q
. Q . .

Process returned 0 (0x0)   execution time : 2.945 s
Press any key to continue.
|
```
